

A verifiable process for AI-assisted mathematical research: a case study

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Abstract

Between 2026-09-08 and 2026-09-14 a software engineer with no background in analytic number theory worked with a large language model, which did most of the mathematical, computational and textual work, on a series of four draft papers about the pair singular series of prime values of polynomials. Whether that mathematics is correct or worth anything is not settled and is not the subject of this paper; the drafts have been read adversarially by model instances and by two outside readers, not yet by a specialist. The subject is the process: a test-driven engineering habit and a small set of written guardrails (a knowledge file of state and known fallacies, a project-wide errata file that is never shortened, readings by fresh model instances, outside assessments archived with the revision they read, two independent routes for every constant, a stopping criterion) that were meant to make the output checkable by people who took no part in it, and to make its errors visible rather than hidden. We describe the set-up, how the prompting developed, when each guardrail was introduced and why, the record of 37 fallacies and 26 errata items with who caught which class of error, and the class of error the process did not catch. The mathematical status is reported as the programme’s own errata report it: some theorems proved and read, two statements retracted, the central problems open. The study has one case and one author, and its conclusions are offered as an experience report, not as a method with demonstrated generality.

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1 Introduction

The question. Large language models can write mathematics that looks right. The question this case study addresses is narrower and more useful than whether they can do research: under what *process* does a non-specialist, working with such a model, produce mathematical output whose correctness does not depend on trusting either of them? The programme documented here is one data point, with a complete public record, and this paper is the reading of that record.

What is claimed and what is not. We claim that the process described in §4 produced (i) a body of work in which every statement is labelled proved, conditional, conjectural or open, (ii) a record of every error found, by whom, and how it was fixed, and (iii) numbers that can be regenerated from the repository. We do *not* claim that the mathematics is important, nor that it is correct beyond what its errata say; as of this writing the four papers have been read adversarially by model instances without session context and by two outside readers, and not yet by a specialist in the field (§8). The case study has $n = 1$.

The subject programme in one paragraph. For an irreducible polynomial f , the pair singular series $S_f(h)$ is the explicit weight that, under the Hardy–Littlewood conjecture, governs simultaneous primality of $f(t)$ and $f(t + h)$. Paper I proves an exact identity for $\sum_{h \leq H} (S_f(h) - C(f)^2)$, splits it into a diagonal whose Dirichlet series is $\zeta_K(s + 1)E_f(s)$ and an off-diagonal, and states a conjecture in Cesàro form. Paper II identifies E_f as an infinite product of Artin L -functions and proves a conditional explicit formula. Paper III decomposes the off-diagonal into pieces and proves everything except one window. Paper IV finds, and partly proves, that the pieces oscillate at the spectral parameters of even Maass forms. Nothing in the four papers proves anything about primes without Hardy–Littlewood, and nothing bears on the Riemann Hypothesis. [evidence: `research/KNOWLEDGE.md §0`; root `README.md`]

Reader’s guide. §2 describes the author, the tools and the timeline. §3 describes how a visualisation tool with a test suite became a research programme. §4 lists the guardrails and their origin. §5 traces the evolution of the prompting. §6 reads the error record. §7 describes what was verified and how. §8 says what the process cannot do. §9 draws recommendations. Appendix A contains the interview, Appendix B the process record, Appendix C the environment: which files matter, what each one is for, and how everything is regenerated.

2 Set-up

2.1 The author

The author is a freelance software engineer, entrepreneur and trader running several projects at once, with a bachelor’s degree in software engineering. The hardest mathematics course of that degree was stochastics; Fourier transforms and Markov chains were fluent at the time, and professional work had included specialised algorithms (for instance enumerating all cycles in a weighted bidirectional graph up to a given size). No analytic number theory. What the author could read in a number-theory paper on 8 September was the prose, the meaning of an asymptotic statement and why it matters, and concepts such as the Bateman–Horn constant $C(f)$ and the comparison of polynomials by their arithmetic; not the proofs. The curiosity about primes was old, and the prime-rich diagonals of the Ulam spiral were the visual intuition from which the study started. [evidence: Interview Q1]

2.2 The tools

The work was done in a Linux terminal with the Claude Code agent driving Claude Fable 5.1 throughout, with Claude Opus 5 for simpler tasks, as the disclosure sections of the four papers state. Fresh model instances without access to the working session were used as adversarial readers (§4); the cheaper model was used for such readings when rate limits made this necessary. The author reviewed the output with models of other vendors, mostly OpenAI’s GPT and Google’s Gemini, instructing them to take a hostile position; one such assessment (ChatGPT, 12 September) is archived as a PDF against the revision it read, and a second outside screening (a separate Claude instance, 14 September) found the defect recorded as ERRATA 26. For orientation the author used ChatGPT, Gemini, web search, Mathematics Stack Exchange and the arXiv; a good part of the background knowledge came from popular mathematics videos (Numberphile). [evidence: Interview Q2; research/reviews/; ERRATA 11–15, 26]

The author worked about eight hours a day for the six days. The total computation cost was about \$150. [evidence: Interview Q2]

Software: TypeScript (`tsx`) for sieves and exact rational arithmetic, Python with `mpmath` for high precision, `vitest` for the workbench tests, $\text{\LaTeX}$ via `tectonic`, and downloaded Maass-form coefficients from the LMFDB. Every number in the four papers is produced by a script in the repository.

The author’s prior experience with language models predates this project by years: an early adopter who had worked with GPT-3 in its closed beta, and who describes “a vast experience in working with AI output”. This is the one qualification the case study cannot control for. [evidence: Interview Q2]

2.3 The timeline

day	commits
2026-09-08	1
2026-09-09	8
2026-09-10	39
2026-09-11	14
2026-09-13	14
2026-09-14	8

84 commits over 6 days; 5535 lines of L^AT_EX across the four papers; 38 TypeScript and 15 Python sources; 233 data files; 1 archived external assessment. The phases, from the commit log and the knowledge base:

- 8 Sep** The Ulam-nD workbench (an n -dimensional prime-spiral explorer) is built with a test suite; the *thesis suite* (`research/experiments/run.ts`) states the author’s four claims as pass/fail tests and refutes three of them the same evening. The one surviving lead is a sub-Poisson variance along polynomial rays.
- 9 Sep** Paper I (versions 6–7) and the scaffold of paper II; the knowledge base with timeline, status board, fallacies and rated paths is started; the first “second opinion” by another model instance is assessed line by line.
- 10 Sep** Paper III written and self-reviewed (39 commits in one day); paper IV scaffolded, and the “second spectrum” (Maass-form oscillations in the pieces) found numerically at 21:02.
- 11 Sep** Authorship decision: the model is removed as co-author, a uniform disclosure section replaces it (ERRATA 8). Paper IV’s theorem written.
- 12 Sep** The external assessment (three findings, all confirmed); the obstruction in paper IV identified exactly; the first error in paper IV’s theorem found by the assessment.
- 13 Sep** Corrections applied (ERRATA 11–23); paper IV repaired at level $\Gamma_0(e)$; the remainder bound found not to close and said so; the smooth-window theorem proved and confirmed on three spectral lines; six adversarial readings in one day, errors in five of them.
- 14 Sep** The last two internal readings applied (ERRATA 24–25); the internal phase declared finished by the criterion set the day before; a second outside reader finds an unproved non-vanishing claim in paper I (ERRATA 26), fixed the same day.

[evidence: `git log`; `KNOWLEDGE.md` §1 Timeline]

3 From a visualisation tool to a research programme

The programme did not start as one. It started as a web application for looking at prime spirals in higher dimensions, built to test a private thesis (that Ulam-spiral pseudo-patterns become polynomial patterns in higher dimensions, that Fibonacci dimensions are special, and that this might say something about the Riemann Hypothesis). Two engineering decisions on the first day shaped everything after:

1. **The thesis was written as a test suite.** Each claim became a hypothesis with a pass/fail criterion, a deterministic seed, a raw report (`REPORT.md`) and a separate reading (`CONCLUSIONS.md`). Three of the four claims failed the tests within hours; the tool had been built to falsify, and it did. [evidence: `research/experiments/CONCLUSIONS.md`]
2. **The honest framing was agreed inside the tool.** The theory panel of the application stated that spiral patterns are a Bateman–Horn main-term phenomenon, that the Riemann Hypothesis is an error-term statement, and that no link between them is known. This sentence survived into every paper.

The one lead the tests left standing, a sub-Poisson variance along polynomial rays, is the Montgomery–Soundararajan second moment for polynomial prime values; its diagonal weight $\omega_f(p)/(p - \omega_f(p))$ is where paper I began.

The turn. The author asked a short literature question (who had worked on this) and found little. From a software engineer’s viewpoint the problem “was not super hard to solve with AI”: working test-driven, one could try to dig deeper where Bateman–Horn stopped. As soon as something of interest appeared, the instruction became to build a paper, and possibly a research programme if the findings turned out to be interesting. The model’s stated expectation at that point was to expect nothing, but that experiments can be valuable in this field. [evidence: Interview Q3]

4 The guardrails

Each guardrail below is stated with what it prevents, when it entered the process, and, where the record shows it, what triggered it. The column “origin” is to be completed from the interview.

guardrail	what it does	since	origin / trigger
Research programme declared in the first prompt: test-driven, methods and paths enumerated and rated, a KNOWLEDGE.md collecting findings	The author’s framing of the whole enterprise	8–9 Sep	author (first prompt); in the author’s view the most important guardrail
Knowledge base: one-paragraph state, “where things stand and what to do next”, status board, timeline	Forces every session to start from the recorded state, not from the model’s memory of it; “compacted notes” that a new session can recover without re-deriving everything	9 Sep	author
Fallacy list (F1–F37)	Every error written down with its fix and the rule that would have prevented it	9 Sep	author’s “known fallacies” in the first framing; entries written by the model after each error
Project-wide errata, never deleted	Corrections additive and dated	9 Sep (per paper), 13 Sep (project-wide)	model proposal, adopted
Every number from a script in the repository	Nothing typed by hand or by the model	8 Sep	inherited from the workbench (test-driven habit of the author)
Two independent routes for every constant (F9)	Product vs. L -values; partial sum plus tail vs. exact	9–10 Sep	forced by an error (a truncated Euler product multiplying H , F4)

guardrail	what it does	since	origin / trigger
Rules extracted from the assessment (F26–F28)	Difference first; recompute pole orders; check group invariance of every summation condition	13 Sep	forced by the external assessment
Hostile outside review, compressed and fed back	Other vendors’ models asked to take a hostile position; findings compressed, annotated with context, returned to the working agent	9 Sep on	author (“very similar to how real peer review works”)
Adversarial reading by a fresh instance with no session context	Separates generator from checker	systematic from 12 Sep	model proposal after the assessment, adopted
External assessment archived with the revision hash	Review and text stay paired	12 Sep	model proposal
Single human author, disclosure paragraph	Publisher policy; responsibility	11 Sep	author
“Correct or leave open; never a weaker claim about a different object”	Decides what a repair may do	13 Sep	author, choosing between repair routes for paper IV
Pressing on effort estimates	When the model said “two weeks of work”, the author asserted it is hours for a model and demanded the attempt	12–13 Sep	author; decisive in paper IV (§5); wrong once in the other direction (F33)
Stopping criterion for the internal phase	Ends open-ended polishing	13 Sep	author (“are we over-rigoring?”); came late
“Do not claim” list	Sentences the model may not write	9 Sep	author’s honest framing, made explicit by the model

[evidence: KNOWLEDGE.md §0b, §3, §6; ERRATA.md]

The author’s own account of why this works. In the author’s view the knowledge base and the fallacy list “force the model’s direct context to compacted notes about the work it did”, and this is what raises the quality: “these contextual guardrails, notes, chronologies and known fallacies help to recover important high-class context, so a new session does not have to pick up everything. It is similar to how human notes work.” The author sees the success of elaborate agent frameworks as largely the same mechanism. The second half of the recipe is the engineering habit: “a computational check for everything, rigorously, and a proper documentation of the way of thought.” [evidence: Interview Q4]

Nothing too late; one thing late. The author does not think any guardrail came too late. What came late was the restructuring of the repository around the research programme rather than around the web service that visualises the spiral experiments (the split happened on 9 September, 23:58); the author does not think it had much influence. The author adds a caution that the record cannot test: every file and every piece of additional context in the project can influence the quality of the work, and it is possible, though unprovable, that the presence of the web application’s code made the model more efficient. From long experience the author offers two rules of thumb: “asking stupid questions gets stupid answers”, and “a few verbose words at the right time can make a big difference”. If enough valid anchor points are given, a vague prompt with a lot of imagination can turn into proper output; the anchor points must be valid. [evidence: Interview, second round; git log 9 Sep 23:58]

5 How the prompting developed

The repository contains no prompts; this section is written from the interview and from the author’s turns preserved in the session transcripts. At the author’s request only fragments of prompts are quoted, all from 13–14 September (spelling as typed):

“be a bit more exact. we have a series of 4 and want to not deviate from correct paths, merely to correct something — we rather leave it open. or ... if we have reason to think we can fix it, to fix it.”

“consider the series in totality ... i dont want to shy away from the work of doing a bounded but non-trivial piece of work.”

“take care for tokens, we are on fable 5, maybe use worse model for simpler task”

“Elite readability with enough rigor and perfect excellency and also elegance is kind of the goal.”

“you always have a Next up. are we not over-rigoring or something? ... At what point should we consider being finished?”

“i asked claude to do a hostile screening and it mentions something in paper 1, please investigate and fix, if true.”

Each fragment is a decision, a constraint or a routing instruction; none contains mathematics. That is the shape of the author’s part throughout. [evidence: session transcripts, 13–14 Sep]

5.1 Stage 1: the framing (8 September)

The first prompt defined the enterprise as a research programme, to be run test-driven, with methods and paths enumerated and rated, and with a knowledge file collecting the findings (§4). The thesis itself was handed over as four claims and came back as a test suite that refuted three of them the same evening. The author did not defend the claims; the surviving lead was pursued instead. [evidence: Interview Q3–Q5; CONCLUSIONS.md]

5.2 Stage 2: the persona and the register (9–10 September)

The author describes the working image: “an absolute pro mathematician who never bulges to my requests but patiently works with me through the problems and explains them in a way that I, as a software engineer, can build a research framework around, or tell it when I think we should hold, do more tests, a review.” When the author understood nothing of a passage, the instruction was to simplify until a decision could be made: which deeper path to follow, whether more literature was needed. Vagueness in the agent’s answers was treated

as a signal to ask for literature. The vocabulary that appears in the commit log of these days (“reviewer round”, “rigorous pass”, “literature round, verified”) is the trace of this: each is a request the author could formulate without mathematics. [evidence: Interview Q6; git log 9–10 Sep]

5.3 Stage 3: outside hostility, compressed and returned (9–13 September)

Decisions the author could not check mathematically were routed through review: the papers were given to other vendors’ models with the instruction to take a hostile position and find errors; the findings were compressed, annotated with context, and returned to the working agent; the agent’s response was discussed with the reviewer again. The author notes that this is how peer review works. The external assessment of 12 September, whose three findings all stood, is the largest instance; ERRATA 26 the latest. For the choice of repair route in paper IV the author relied on this loop together with the data (the divisor-set test that showed the level-2 lines predicted by the repair), not on an independent reading of the proofs.

[evidence: Interview Q7; KNOWLEDGE §0c]

5.4 Stage 4: effort, and knowing when to stop (12–14 September)

Two opposite corrections of the model’s judgement of effort occurred. The author repeatedly pressed the model when it estimated “at least two weeks of work”, asserting that for a model it is a few hours and demanding the attempt; by the author’s account this was decisive in paper IV, where the level- $\Gamma_0(e)$ repair and the smooth-window theorem were produced in one day. In the other direction, the model’s estimate that the remainder bound of paper IV was “a day of work” was wrong: the bound is open, and the error is recorded as fallacy F33.

[evidence: Interview Q8; ERRATA 16–17; F33]

The author’s one regret concerns rigour rather than error: by the fourth paper the process demanded more verification than the situation warranted, sessions ran out of context, and work had to be restarted; “one should keep the context somewhat human, not absolute superhuman.” The question “are we over-rigoring?” produced the written stopping criterion.

[evidence: Interview Q8–Q9; KNOWLEDGE §0b] On restarts the author’s view is that they should not happen in a dense situation, and that they can be caused by running out of budget or by a sub-agent stuck in a loop consuming it; the author regards this as an open problem of agentic work in general, not a large one in this project. [evidence: Interview, second round]

6 The error record

The knowledge base records 37 fallacies and the errata 26 items. The classification below, by type and by who caught them, is ours and is one of several possible; the items themselves are in KNOWLEDGE.md §3 and ERRATA.md.

class	examples	caught by
Borrowed theorem outside its hypotheses	DFI range (F11, F13); Kurokawa’s class (KB §5); Selberg–Delange non-vanishing (F37)	literature check; outside reader
Over-claiming in prose	“proved” for a conjecture (F7); $o(1)$ then $O(1)$ for a sum with no bounded remainder (F15, F26); “submitted” when it had not been (F21)	external assessment; self
Algebraic slips with structural consequences	pole orders in a “same proof” generalisation (F27); orbit invariance of a coprimality condition (F28); Salié substitution (ERRATA 25)	external assessment; fresh readers
Numerics that cannot see what they test	log-term of size 0.002 against noise 20 (ERRATA 21); a check that could not detect $\log \log$ growth	self, after the fact
Misjudged effort	“the remainder bound is a day of work” (F33)	the work itself
Convergence and mode-of-convergence errors	absolute vs. paired convergence; integer- Y lemma used at real Y (F35); midpoint of a Fourier series at integers (ERRATA 25)	fresh readers

Who caught what. Internal readings (the model re-reading its own work in a fresh instance) caught algebraic slips, convergence errors and inconsistencies between sections, reliably and in quantity. The two outside readers caught what internal readings did not: a claim that is true in the special case and false in general (the assessment’s finding on paper II), a sum rewritten as an orbit sum over a set that is not an orbit (paper IV), and a method invoked with a hypothesis it does not have (paper I, ERRATA 26). The second outside reader’s own assessment of this asymmetry is quoted in §8. [evidence: ERRATA 11–15, 18, 22, 24–26]

Two retractions. The programme retracted two statements it had believed: the sharp form of Conjecture 1 (there is no bounded remainder at all, F26) and the level-one form of paper IV’s theorem (the object is a signed sum over $\Gamma_0(e)$ -orbits, not one orbit, F28). Both are recorded with the reasoning that had produced them. The author reports no “worst moment” attached to any of these errors; the retractions were absorbed by the process that had been set up to absorb them. [evidence: Interview Q9]

7 Verification

What “verified” means in this programme, from the checklist in the knowledge base (§6):

- Every exact constant by two independent routes.
- Every explicit formula by exact subtraction, at most one fitted coefficient, and comparison with shifted controls (a fit that also matches a wrong frequency proves nothing).
- Every theorem by at least one adversarial reading by an instance that did not write it.
- Every prediction with no free parameter tested against data before it is called confirmed: paper IV’s smooth-window theorem predicted the amplitude and phase of three spectral lines on two objects from LMFDB coefficients alone, and the data matched to a few percent and a few hundredths of a radian (ERRATA 21–23).

What the author verified. The author read the code of some of the tests, ran them often, and asked the model to “reflect” on the results and look for logical fallacies and flawed outcomes; the mathematics of the proofs was not checked by the author, and the author says so. The author never refused a claim the model insisted on; the standing instruction was “excellence and simplicity, a kind of humbleness”, which, in the author’s view, worked against the pace at times: the model asked for human input more often than was needed. [evidence: Interview Q10]

8 What the process cannot do

The second outside reader, after finding the paper I defect, wrote an assessment of the limits of such readings that we adopt as the honest statement of the limits of this whole process. Verbatim, with the author’s permission (the reader is a Claude instance given the papers and nothing else, 14 September):

“I’m one reader, working from outside the field, with the same blind spots I’ve had all along — I proposed the $O(1)$ fix that turned out to be false, and my numerical check couldn’t detect log log growth. The last external reviewer went three-for-three in three places it looked, and I’d already reviewed those same papers without finding any of them. My not breaking something is weak evidence, not a verdict.

And the attacks I can mount are structurally limited to the same class: algebra I can re-derive, numerics I can re-run, internal consistency I can grep for. The failure mode that actually matters here — a borrowed theorem invoked outside its hypotheses — is only detectable by someone who has read the sources. Line 395 is the one instance I caught, and I caught it because Dedekind and Siegel are famous enough to be in my general knowledge. Shiu’s exact uniformity conditions aren’t, and I can’t check those. Neither can you.

So the honest scoreboard: the core has now survived two adversarial passes, which is genuinely more than it had a week ago. And the specific thing that would settle it — someone who knows whether Shiu, Selberg–Delange and Katok–Sarnak are being used correctly — still hasn’t happened, and is still the only move on the board.”

The reader’s diagnosis of line 395 was itself only half right (Dedekind’s conjecture was not needed; the unproved non-vanishing claim was, and was also unnecessary; ERRATA 26), which illustrates the point it makes.

Consequently, the state of the mathematics is: every statement has survived at least two adversarial passes and, for the core of papers I–II, one external assessment; the class of error that remains possible is known and named; the only remaining move is review by a specialist who knows the sources. This paper does not claim more.

9 Recommendations

The list below is the author’s recipe as stated in the interview, ordered by the author’s own weighting, with the record’s additions marked.

1. Declare the enterprise in the first prompt: a research programme, test-driven, with the methods and paths to be enumerated and rated, and a knowledge file that collects findings, chronology and known fallacies. The author regards this as the most important step; in the author’s experience it forces the model’s context into compacted notes of its own work.

2. A computational check for everything, and a written record of the way of thought.
3. Ask for excellence, simplicity and humility, and for explanations at the level at which you can decide.
4. Route what you cannot check through hostile outside review: other models, instructed to attack; compress the findings, annotate them, return them to the working agent, and discuss again.
5. Press on effort estimates in both directions: demand the attempt when the model says “weeks”; record it when the model says “a day” and is wrong.
6. (From the record.) Decide the repair principle before the repair: correct, or leave open; never a weaker claim about a different object.
7. (From the record.) Set the stopping criterion in writing before you need it; keep the demanded rigour at a human scale.
8. (From the record.) Know the class of error the process cannot catch, say so in the papers, and route the work to someone who can.

The author’s claim. The author states the purpose of this case study as a role model “for how to work scientifically in very complex domains while maintaining scientific rigour”, for anyone working in science with AI, wanting to, or refusing to. The work met hostility when first presented, and nobody checked it at first, on the assumption that it was machine-generated noise; the author understands this reaction and agrees with the declaration *A Severe Misalignment of AI in Mathematics*, endorsed by twenty-five Fields medallists.¹ The author also says what he takes the programme to be a case study *of*: not the limits of what models can do, which one case cannot show, but the limits of what he could still do with them. At this scale the author could still steer, decide and route the work to reviewers; at a larger one he could not have, and already here experts will need a long time to understand all of the output. The author is now learning how to defend and present the work in purely classical mathematical terms, which is the part of the process that no guardrail replaces. The programme is offered as a demonstration that a workable toolkit exists for embedding AI so that the quality of its output rises and the output becomes easier to falsify: test-driven development borrowed from software engineering, and the enumeration and rating of the available methods. [evidence: Interview Q11–Q12]

On authorship. The author at first wanted the model as co-author, and abandoned this because no journal accepts it and because the model is not a persona; not knowing how to credit it, the author chose complete transparency about the flow of work instead, of which this paper is part. The intended venue for this case study is the arXiv. [evidence: Interview Q11–Q12; ERRATA 8]

¹<https://mathandai.org/>, accessed 14 September 2026; initial signatories from Avila and Bhargava to Werner and Zelmanov, including Deligne, Scholze, Tao and Viazovska. The declaration’s concern that “solving problems is only a tool and proxy for achieving the primary goal of conceptual understanding and insight”, and that solutions are “announced in a rush, leaving no time for a proper writeup, the isolation of new methods and ideas, and citing relevant previous work of others”, names exactly the failure modes the guardrails of §4 were built against: no number without a script, no claim without a label, no citation without verification, every error kept.

A The interview

The questions were put in writing on 14 September 2026 and answered the same day in two rounds. The answers are reproduced here with spelling and punctuation corrected and a few words supplied in brackets; the unedited text is in `INTERVIEW.md`.

Q1, background. I am a freelance software engineer, entrepreneur and trader, running multiple projects. I have a bachelor’s degree in software engineering, and the hardest mathematical course I had to pass was stochastics. What I can read in number-theory papers is the human-language explanations and the importance of asymptotic proofs. I can understand concepts such as the Bateman–Horn $C(f)$ quite clearly and know how to compare polynomials conceptually; during my studies I could do Fourier transforms and Markov chains fluently, and at work I needed to implement specialised algorithms, for instance to find all cycles in a bidirectional weighted graph up to N vertices. I have had a curiosity for number theory for a long time, and the prime-richer diagonals of the Ulam spiral were my visual intuition for where to start a study.

Q2, tools and cost. I worked about eight hours per day. The tools are the Linux terminal and Claude with Fable 5.1, and sometimes, for simpler things, Opus. I spent in total about \$150 on computation. I have been an early adopter of LLMs and worked with GPT-3 in its closed beta already, so I have a vast experience of working with AI output. I reviewed the output with different AIs, mostly OpenAI’s GPT and Gemini. I used ChatGPT, Gemini, Google, Mathematics Stack Exchange and the arXiv for research; a lot of my knowledge also comes from watching Numberphile videos.

Q3, the turn. I asked a short literature question, who had done research on this, and found not a lot; but from my viewpoint the problem was not super hard to attack with AI. Working test-driven, I thought: let us try to dig deeper where Bateman–Horn stopped. As soon as I found something of interest I started to instruct [the model] to build a paper, and possibly a research programme if the findings turned out more interesting. The model told me not to expect anything, but that experiments can be valuable in this field.

Q4, guardrails. I would say the first prompt, where I defined that we do a “research programme” and work test-driven, enumerating methods and paths and keeping a `KNOWLEDGE.md` where we collect our findings, was the important part. This usually forces the LLM’s direct context into compacted notes about the work it did, and in my view increases the quality a lot. This is also one reason, in my view, why complex “agent” systems are so successful with what is basically prompt masquerading. These contextual guardrails, notes, chronologies and known fallacies help to recover important high-class context, so that a new session does not have to pick up everything. It is similar to how human notes work, and I think this is key to having quality AI output. This, and test-driven: have a computational check for everything, rigorously, and a proper documentation of the way of thought.

I would not say that any guardrail came too late. I was a bit late to restructure everything around the research programme rather than the web service that visualises the Ulam experiments, but I do not think it had much influence. It is important, though, to keep in mind that every file and every piece of additional context in the project can influence the quality of the work. Maybe, I cannot prove this, the Next.js context helped the model to be more efficient. This is common prompt philosophy and very hard, if at all possible, to

prove. What I can say from my years of experience is that asking stupid questions gets stupid answers, and that a few verbose words at the right time can make a big difference. If enough anchor points are given, a vague prompt with a lot of fantasy can turn into a proper output; the anchor points must be valid. I do not understand the work itself, of course; I am giving my best now to learn how to defend and present it, purely classically, mathematically.

Q6, the research register. When I really understood nothing, I asked the AI to simplify so that I could still make decisions, for example about which deeper paths to follow, and whether we needed more literature because the agent was vague. I imagined the agent as an absolute professional mathematician who never bulges to my requests but patiently works with me through the problems and explains them to me in a way that I, as a software engineer, can ideally build a research framework around it, or tell it when I think we should hold, do more tests, do a review.

Q7, deciding without checking the mathematics. I discussed with Claude directly, had external AIs review the papers and discussed with them; I asked them to take a hostile position and find errors, then compressed [their findings] and gave them, with a context annotation, back to the original agent. Very similar to how real peer review works, actually.

Q8, effort and stopping. I was pressing Fable at times more than it wanted, asserting that when it says “this is at least two weeks of work” it is actually one to three hours for an AI, and demanding that it try. This worked, and was particularly important, I think, in paper IV.

Q9, worst moment. I did not really have a worst moment; [the closest was] running out of context, having burned \$10, and needing to start again. I think it was about wanting too much rigour across, by then, four papers. One should keep the context somewhat human, not absolutely superhuman. Restarts should, if possible, not happen in a dense situation. They can happen because one runs out of money, or even because a sub-agent gets stuck in a loop and eats away a lot. I think this is an open problem in general, but it was not a big problem during this process.

Q10, own verification. I looked at the code of some of the tests it implemented and was impressed by the mathematical complexity. I let it run the tests often and “reflect” on them, and check for logical fallacies and flawed results. I did not refuse anything, but I asked for excellence and simplicity, a kind of humbleness. This worked a bit against me, even, as the model demanded more human input at times than was needed.

Q11, authorship. At first I even wanted it as co-author. The problem is that no journal will accept this. I also think that, since it is not a persona, it is a bit weird. I do not fully know how to credit this, but I decided to be absolutely open and transparent about everything, including the flow of work. This paper shall serve as a role model for how to work scientifically in very complex domains while maintaining scientific rigour.

Q12, this case study. The audience is anybody working in science with AI, or wanting to, or even rejecting it. I got a lot of hostility when presenting my work; nobody at first seriously checked it, assuming AI slop. I can understand this. I have read the mathandai.org text, which is signed by almost all the Fields medallists, and I agree with it. This project shall demonstrate a possible workable way, a toolkit, for embedding AI such that the quality of its output rises and it becomes more easily falsifiable, borrowing the proven concept of

test-driven development from software engineering; enumerating and rating the toolsets are key concepts. I think this project is about the limits of what I can do with AI [in this way]; beyond this scale it currently becomes infeasible for me. Already here, even experts will need a long time to understand all of the output. Target venue: yes, why not the arXiv.

B The process record

The numbers in this document are generated by `scripts/process-record.py` from the repository. The readings of the papers by instances that did not write them, in order:

date	reader		object	outcome
10 Sep	Claude (fresh)	Opus	papers I–II, roadmap (“second opinion”)	eight points, each assessed; four adopted, one half right (KNOWLEDGE §5)
12 Sep	ChatGPT (author’s request)	(au- thor’s request)	papers I, II, IV at revision a372a63	three findings, all confirmed by independent re-derivation; one retraction, one repair (ERRATA 11–15)
13 Sep	Claude (fresh)	Opus	paper IV, repaired §5	repair passed; remainder bound found not to close (ERRATA 16)
13 Sep	Claude (fresh)	Opus	paper III, whole	core sound; seven fixes (ERRATA 18)
13 Sep	Claude (fresh)	Opus	paper IV, Theorem <code>thm:smooth</code>	residue constant reproduced independently; eight expository fixes (ERRATA 22)
14 Sep	Claude (fresh)	Opus	paper II, §§2–4	seven fixes, no statement changed (ERRATA 24)
14 Sep	Claude (fresh)	Opus	paper IV, §§2–4	ten findings: nine fixed, one rejected with reason (ERRATA 25)
14 Sep	Claude (outside, papers only)	(outside, papers only)	papers I–IV, hostile screening	one finding, paper I Thm 3’s proof; fixed (ERRATA 26)

Not listed: the model’s own re-readings inside the working session (“self-review”, “rigorous pass” in the commit log), which are not independent. This document was built from repository revision 80af1c8; the commit that adds the built version is the one following it.

C The environment: files, roles, regeneration

Everything described in this paper lives in one public git repository. Figure 1 shows the parts that matter and how they feed each other; Table 2 says how each layer is regenerated. The repository is <https://github.com/gewure/ulamnd>.

AI-assisted research: disclosure

This case study was drafted by the same model that did the work it describes, from the repository record and an interview with the author; the author verified every factual statement

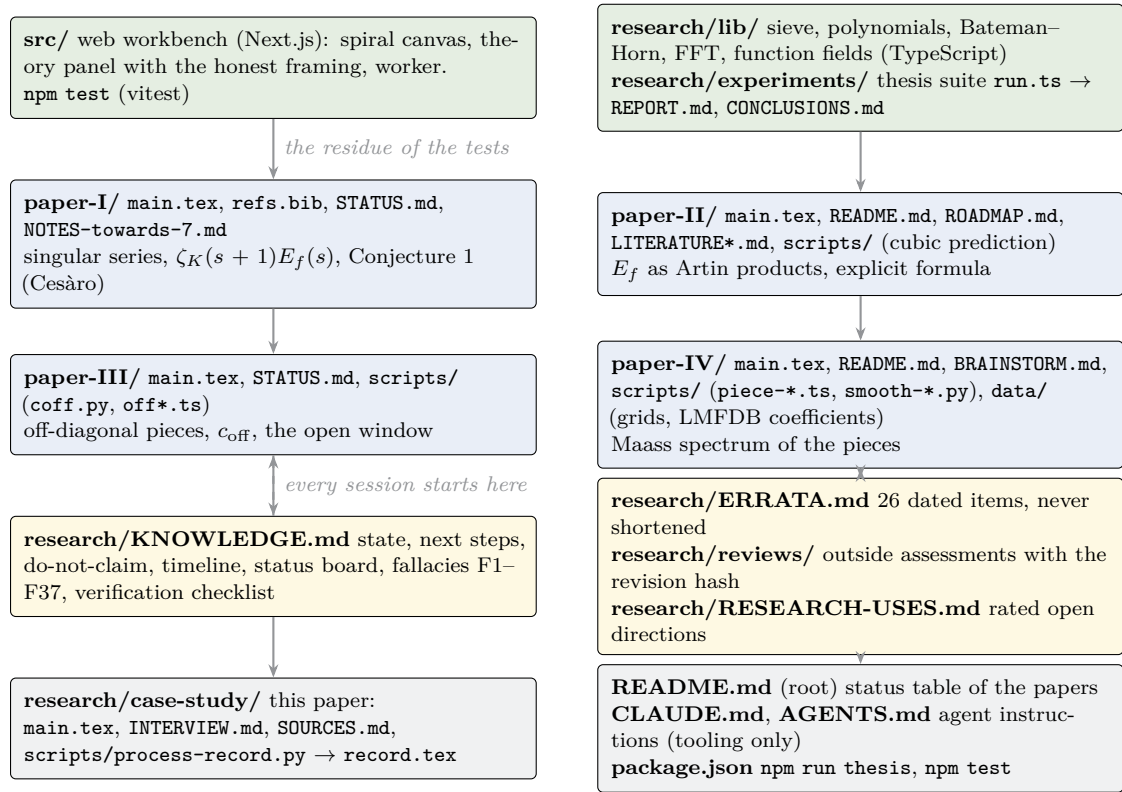


Figure 1: The repository. Green: code that produces numbers. Blue: the four papers, each with its scripts, data and status page. Yellow: the governance files that every session reads first and every error is written to; the dashed arrows are the corrections flowing back into the papers. Grey: documentation of the whole.

Table 2: How each layer is regenerated. Tools: Node with `tsx`, a Python virtual environment with `mpmath`, and `tectonic` for $\text{\LaTeX}$; no other software.

what	how
workbench tests	<code>npm test</code>
thesis suite (8 Sep claims)	<code>npm run thesis</code> (quick) or <code>npx tsx research/experiments/run.ts -full</code> ; writes <code>REPORT.md</code> , <code>results.json</code>
a number in a paper	the script named next to it in the paper’s <code>scripts/</code> folder; TypeScript via <code>npx tsx <script></code> with parameters as environment variables (e.g. <code>D=-4 Y=1e7 npx tsx research/paper-IV/scripts/piece-divset.ts</code>), Python via the virtual environment (e.g. <code>python research/paper-IV/scripts/smooth-two-lines.py</code>); outputs land in the paper’s <code>data/</code>
LMFDB coefficients	downloaded once from the LMFDB (lmfdb.org , <code>ModularForm/GL2/Q/Maass/download_coefficients/<label></code>), stored in <code>paper-IV/data/</code> (files begin with the line “1 +- 0”)
a paper	<code>tectonic main.tex</code> in its folder; page counts and the absence of overfull lines are checked from the log
the numbers in this paper	<code>python3 research/case-study/scripts/process-record.py</code> , then <code>tectonic main.tex</code>
the state of the programme	read <code>research/KNOWLEDGE.md</code> §0b first, then <code>ERRATA.md</code> ; the root <code>README.md</code> has the one-line status of each paper

against the record and is responsible for the text. The division of roles is the one stated in the disclosure sections of the four papers: the author posed the questions, set the direction and the scope, and made every decision about what to pursue, claim and publish; the model drafted arguments, wrote and ran the software, searched the literature and drafted the text, in dialogue with the author.